Learned Remote Proprioception and Contact Force Estimation from Motor and Load Cell Signals

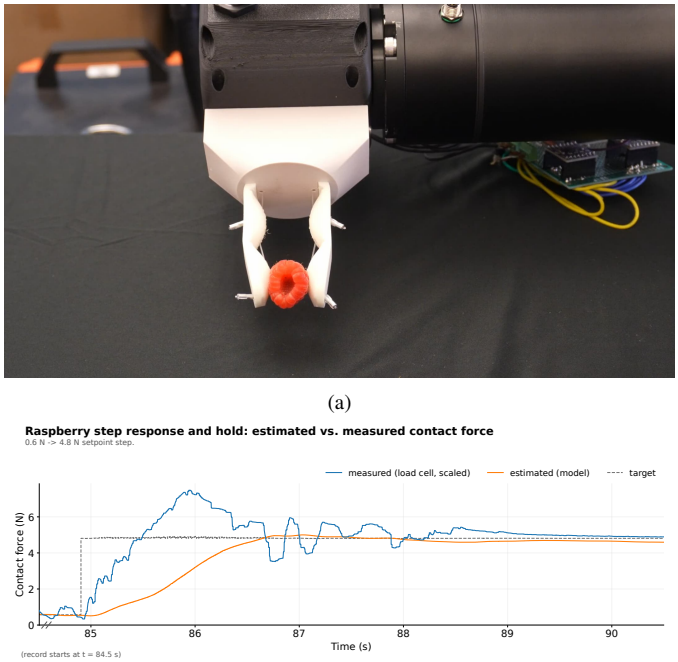
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***Abstract*—Soft robotic hands offer compliant, human-safe interaction but remain limited by the difficulty of obtaining reliable proprioceptive feedback for closed-loop control without introducing sensing hardware that constrains the hand's passive deformability and natural compliance. We extend a remote sensing and learning approach to achieve proprioception and contact force estimation in multi-tendon, multi-degree-of-freedom soft fingers that require no embedded electronics or external visual tracking. This method relies exclusively on actuation-side telemetry, combining motor position, velocity, and drive current with independent tendon tension measured via load cells. We train a shared-trunk gated recurrent unit (GRU) network on this data to predict joint angles, contact force magnitude, and contact onset. Through evaluation across various loading regimes, we demonstrate that dedicated tendon load cells improve force-structure tracking, resolving true kinetic magnitudes once actuators mechanically stall, allowing the network to achieve a 4.11° joint mean absolute error (MAE), a 0.34 N contact-force MAE, and a 0.79 contact-onset F1 score under standard physical interactions. Finally, we successfully deploy this framework for real-time closed-loop force regulation at a 90 Hz inference rate, achieving highly stable steady-state tracking with a 0.32 N MAE, advancing the feasibility of robust, sensorless soft end-effectors in unstructured environments.**

I. Introduction

Dexterous robotic hands are conventionally designed around rigid links connected by discrete joints, making their kinematic configurations highly deterministic and easily observable via joint-space encoders. More recently, soft robotic hands have emerged as a promising alternative, offering advantages in material compliance, structural durability, and human-safe interaction. Fabricated primarily from elastomeric polymers, soft hands are robust against impact fractures, cost-effective to manufacture via additive fabrication techniques, and safe when interacting with fragile, unstructured environments. However, harnessing the full potential of soft robotic hands for dexterous manipulation demands closing the control loop. Taking inspiration from human manipulation, which relies on a rich, multi-modal sensory network, a soft hand requires both joint-level proprioception and contact-force awareness to regulate physical interactions. Achieving this coupled sense of proprioception and force feedback is the objective of this work.

Proprioception and tactile feedback are conventionally obtained by embedding discrete encoders and tactile skins directly within the joints and fingertips. However, embedding rigid electronics into compliant finger bodies compromises the advantages of soft robotics for two primary reasons. First, it degrades the hand's mechanical durability and low-cost nature.



(a)

(b)

Fig. 1: Learned remote proprioception and real-time force control: (a) Hardware setup during manipulation with a delicate object (Raspberry), relying solely on actuation-side sensing; and (b) Closed-loop step response (0.6 N $\rightarrow$ 4.8 N) showing the filtered model estimate driving the controller alongside ground-truth load-cell verification.

Soft appendages are typically simple, mechanically robust, with minimal failure modes and low cost. Second, embedding rigid elements restricts local material deformation, thereby compromising the hand's compliance and safe interaction capabilities.

Real-world manipulation tasks demand multi-degree-of-freedom fingers with multi-digit coordination. This work advances actuation-side sensing for tendon-driven soft robotic hands towards real-world deployment, making the following key contributions:

- Scales actuation-side proprioception and contact force learning to coupled, multi-tendon finger kinematics.
- Evaluates the learned models across unseen physical interaction scenarios.

- Deploys the framework for real-time, closed-loop contact force regulation.

II. Related Work

In soft robotics, various sensorization strategies have been used but no single approach has become standard due to the challenges of instrumenting deformable materials. This differs from the traditional rigid robotics, where precise motor encoders and standalone force sensors are easy to integrate and are widely adopted. These include:

A. Actuation-Side Sensing

Spiers et al. [1] leverage temporal current signatures and tracking errors from variable-friction robotic fingers to perform object classification and geometry identification during within-hand manipulation. Bauer et al. [2] introduce a remote sensing method, where a model is trained on motor and load cell data to estimate joint angles, contact force, and contact location for a TPU-printed single-tendon finger. Sadati et al. [3] apply the Cosserat rod model with information from a proximal 6-axis load sensor to image soft tissue using a 3-degree-of-freedom continuum appendage.

B. Fluidic & Barometric Sensing

Gilday et al. [4] introduce a conformable barometric sensor skin allowing force measurement across varied end-effector geometries. Tawk et al. [5] present a 3D-printed soft monolithic finger which houses pneumatic sensing chambers (PSCs) acting as position and touch sensors. Hyperelastic material and finite element models are then used to relate the change of volume in the PSCs to the corresponding mechanical input. Lin et al. [6] show how fiber-based displacement sensors can measure bending, torsion, and elongation to reconstruct diverse finger joint states.

C. Optical & Vision-Based Tactile Sensing

She et al. [7] integrate embedded cameras with convolutional neural networks, enabling rich tactile sensing by tracking internal surface deformation via axially positioned markers and reflective coatings. Liu et al. [8] showcase a vision-based soft polyhedral network that infers 6-axis contact forces and torques from visually observed structural deformations, incorporating a creep and relaxation model to compensate for material viscoelastic hysteresis.

D. Material-Based Transduction

Roberts et al. [9] demonstrate resistive and piezoresistive sensing by measuring changes in electrical resistance when materials like conductive elastic rubber deform. For proximity and contact detection, Taylor et al. [10] utilize capacitive sensors to measure changes in electric fields generated around the hand's soft material. Furthermore, Roberts et al. [9] and Weiner et al. [11] capture multi-axis forces using magnetic sensing, which relies on embedding magnetized microparticles or discrete magnets into elastomers to detect changes in the internal magnetic field during deformation.

E. Alternative & multi-modal approaches

Puhlmann et al. [12] utilize acoustic sensing within the RBO Hand 3, leveraging sound propagation through soft materials to infer contact location, intensity, and other interaction states. Bauer et al. [2], discussed above, also embed inertial measurement units (IMUs) and accelerometers directly into the fingertips of tendon-driven soft fingers, focusing specifically on vibration and slip detection. Addressing pre-contact awareness, Weiner et al. [11] incorporate time-of-flight (ToF) distance sensors within robotic and prosthetic fingertips to determine the proximity from the finger to the target object before contact occurs.

III. Hardware and Sensor Setup

A. Design and Fabrication

The soft tendon-driven fingers in this work are 3D printed from Thermoplastic Polyurethane (TPU) using standard fused-deposition modeling (FDM) printers (Filaflex Foamy 1.75 mm filament, Bambu Lab H2D). Each soft finger features two joints, formed by material creases. Tendons are routed internally through printed channels, with one tendon actuating the proximal joint and a second tendon actuating the distal joint. As shown in previous works, the deformation behavior at the joints can be represented using rigid-body dynamics approximation [13]. In a similar setup to previous work [2], the load cell is placed in the wrist of the hand near the motor, leaving the finger/hand completely sensorless as shown in Figure 2.

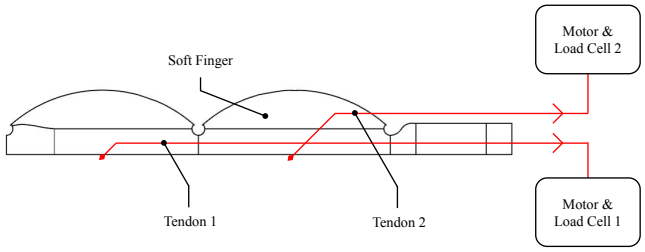


Fig. 2: Tendon and sensor layout of the multiple-degree-of-freedom soft robotic finger with motor and load cell remote sensing capabilities.

B. Sensorization Strategy

The sensor setup measures both the actuation state of the finger and the tension applied through each independently driven tendon. All primary sensors are placed at the motor level, remote from the finger body, so that the structure remains entirely sensor-free and its compliance is unaffected by embedded electronics. An external load cell is used to obtain ground-truth contact force.

- **Actuation-side sensing**: Tendons are actuated by Dynamixel XC330-T181-T motors, one per tendon. Each motor reports encoder position, velocity, commanded position, and drive current. Motor position provides tendon



(a)



(b)

Fig. 3: Experimental setup for ground-truth data collection: (a) Vicon markers are fixed to the left finger, utilizing three markers per link in a non-planar configuration to capture rigid-body transformations. Data is collected exclusively from this instrumented finger to train the model, which is subsequently deployed zero-shot to the uninstrumented right finger; and (b) Handheld force probe equipped with an S-type load cell and a tip tracking marker to record synchronized contact wrench and interaction locations during poking trials.

displacement at the actuator, while current serves as a coarse proxy for actuation effort.

- **Tendon Tension Sensing:** Tendon tension is measured independently on each actuation line using a DYLY-109 S-type load cell, digitized by a Texas Instruments ADS1220 24-bit ADC, and processed by an ESP32-WROOM microcontroller that streams the data via ROS 2.
- **Ground-truth contact force sensing**: To obtain ground-truth contact force data, a DYLY-108 S-Type load cell mounted on a handheld probe is used during the poking procedure adapted from previous work [2], as described in IV-B.

IV. Method

A. Testbed Setup

To capture ground-truth motion of the multi-degree-of-freedom finger, we use a Vicon motion-capture system operating at 120 Hz. The two-link finger is instrumented with six markers—three mounted non-planarly on each link to define independent 6D reference frames—alongside eight reference markers on the static wrist, yielding 14 tracking markers in total (Figure 3). Tracking each link independently ensures that local occlusions do not disrupt joint-angle estimation. In addition, three static markers on the workspace surface define the ground plane, while an additional marker on the force probe tip registers contact locations.

Prior to assembly into the wrist, each load cell was calibrated using a 100 g weight.

B. Data Collection

We ran three MOCAP sessions, yielding three recorded datasets: a no-contact set comprising unconstrained free-space motion, a contact set in which the finger was pressed against a force probe, and a contact set where large force was applied to the finger via the force probe for hyperextension. All datasets span the same movement primitives. Table I details the specifics of each movement with the number of trials per dataset.

TABLE I: Description of Recorded Movements per Dataset

Movement	Description	Trials Per Dataset
Flex_Hold_100	The primary flexion joint is actuated to a target amplitude (100% or 80%) and held at a fixed angle for the duration of the repetition.	3
Flex_Hold_80		3
Hinge_Hold_100	The hinge joint is actuated to a target amplitude (100% or 80%) and held at a fixed angle.	3
Hinge_Hold_80		3
Flex_Hinge_Async_100	Asynchronous, sequential actuation of the flexion and hinge joints.	3
Flex_Hinge_Sync_100	Synchronous, simultaneous actuation of the flexion and hinge joints.	3
Random_Poses	The finger joints are driven through a series of randomly generated coordinate combinations to capture a broad, generalized distribution of joint states.	9

Prior to data collection, we used a manual calibration procedure to establish the joint's neutral (zero) and fully actuated states by incrementally spooling each tendon via keyboard commands. The zero position was defined at the exact transition point between tendon slack and tension. The fully actuated position was set by driving the actuator until the finger met its mechanical hard stop without excessive tendon strain.

Before each trial, tendon pretension was reset by hyperextending the finger, after which all load cells were tared. To temporally synchronize the telemetry and optical tracking streams, we performed a ground-tap event with the force probe, aligning the transient load-cell contact spike with the probe marker's impact trajectory ($Z = 0$) in the motion-capture stream.

In the first experiment, the contact probe was secured in a fixed position. As the finger actuated through its predefined trajectories, it pressed against the stationary probe, simulating the reactionary forces of interacting with a rigid object in the workspace. For all recording takes in this phase, contact was restricted exclusively to the distal link. To complement this object-interaction data, a second dataset of contact data was collected to capture the physical limits of the hardware. During

this secondary phase, large external forces were applied to the finger to intentionally induce joint hyperextension. This approach was designed to record the extreme loading conditions the finger might encounter during actual operation, ensuring the resulting model captures the full spectrum of potential contact events and can effectively interpolate between them. In total, there were 3 datasets of what is displayed in Table I: one no-contact, a normal operating contact dataset, and an extreme contact dataset.

C. Data Processing

Hardware telemetry, including load cell and motor data, was extracted directly from the recorded ROS bags. To derive the joint angles from the MOCAP system, the 3D marker positions were processed using the Kabsch algorithm via Singular Value Decomposition (SVD) [14], [15]. For each rigid link, the algorithm computed the optimal rotation and translation by minimizing the Root Mean Square Deviation (RMSD) between the instantaneous marker clusters and a pre-defined Unified Robot Description Format (URDF) reference. Specifically, each point set is centered at its centroid to decouple translation from rotation, after which SVD was applied to the cross-covariance matrix of the centered points to extract the optimal rotation matrix, including a reflection check to ensure a proper rotation. The corresponding translation was then recovered from the aligned centroids. The relative transformations between adjacent links were then used to compute the joint angles.

This approach was chosen over traditional Inverse Kinematics (IK), which relies on solving for joint angles across an entire continuous kinematic chain. By using the Kabsch algorithm, each link can be solved for individually, preventing the compounding of errors caused by non-ideal, off-axis joint deformations under physical load.

Following the processing of the hardware and MOCAP streams, the two systems were temporally aligned using the ground-tap synchronization event. The transient spike in the ROS load cell data was matched to the corresponding geometric contact event ($Z = 0$) in the MOCAP data. Using this temporal offset, the data streams were shifted, interpolated to a common sampling frequency of 120 Hz, and merged into a single dataset for each trial.

D. Model Architecture and Training

A model architecture similar to previous work [2] was used and is shown in Figure 4, in which a shared two-layer gated recurrent unit (GRU) trunk, configured with a hidden size of 64 and dropout, feeds three task-specific multi-layer perceptron (MLP) heads. The network operates on a feature vector per timestep comprising 2×4 motor signals (motor position, velocity, current, and command) alongside 2 load cell signals and time. The model infers three outputs: joint angle, contact force magnitude, and contact onset.

A shared GRU trunk was chosen over three independent per-task GRUs because the physical states of the soft finger: kinematics (posture) and kinetics (contact forces) are inherently coupled, and tendon tension is influenced by both the finger's internal joint angles and any external forces on the finger. By utilizing a shared hidden state, the network leverages this mutual information as a multi-task regularizer to prevent overfitting to isolated tasks. Furthermore, a unified trunk reduces the model's parameter count and inference latency compared to three independent GRUs, which is paramount for real-time closed-loop control.

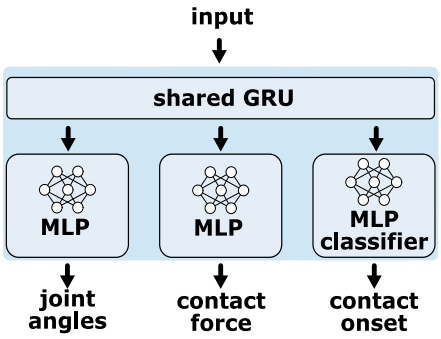


Fig. 4: Model Architecture: 2-layer shared Gated Recurrent Unit (GRU) trunk with individual Multi-Layer Perceptron (MLP) heads for joint angles, contact force, and contact onset prediction.

The GRU trunk operates on a fixed-length input window of $T = 300$ timesteps, which corresponds to a look-back period of 2.5 s for input data streaming at 120 Hz. Each of the three MLP heads features a LayerNorm–GELU–Dropout block applied to the GRU trunk output, producing per-task predictions.

To optimize the network, a composite loss function was applied, utilizing distinct metrics tailored to each specific output: Masked L1 loss (MAE) for the continuous joint angle head, Binary Cross-Entropy with Logits (BCEWithLogitsLoss) for the binary contact onset head, and Masked Smooth L1 (Huber) loss for the continuous contact force head. The total loss is computed by $\text{loss}_{total} = \text{loss}_{joint} + 0.5 \times \text{loss}_{onset} + 1.0 \times \text{loss}_{force}$.

Training utilized the Adam optimizer with a learning rate of 4×10^{-4} and a batch size of 1024. Models were trained on the NVIDIA Jetson Thor for a maximum of 80 epochs with early stopping (patience of 15) and best-checkpoint selection based on validation loss. To stabilize inference performance during live deployment, an exponential moving average (EMA) of the model weights (Polyak averaging) was maintained throughout training using a decay rate of 0.999.

V. Results

Models were trained using a fixed split evaluation method, with takes either being kept as training or test takes. Three takes are held out per dataset for testing, while the rest are used for training. Model T is evaluated on the three held-out no-contact takes, using the rest of the no-contact takes for training. Model Q is evaluated only on contact takes, using

three standard and three extreme-contact takes, and trains on the remaining 75 takes, which include all 27 no-contact takes.

Using the same model architecture, we trained two primary models: Model T, a no-contact model that was trained on unconstrained motion to predict joint kinematics for pre-grasp positioning and collision avoidance, and Model Q, a contact model, trained to simultaneously predict joint kinematics, contact forces, and contact states during object interaction. The quantitative results are summarized in Table II where normalized joint error is joint MAE divided by the joint's full actuated range. In addition, we train motor-only counterparts, Models U and V, which use the same architecture and splits but omit the load cell channels from the training data. These are analyzed in Section VII.

TABLE II: Model Performance Across Operating Regimes

Model	Evaluation Set	Joint MAE (°)	Norm. Joint Error (%)	Force MAE (N)	Force ρ	Contact F1
T (No-Contact)	3 Free-Space Takes	2.24	4.21	–	–	–
Q (Contact)	3 Standard Takes	4.11	6.71	0.34	0.73	0.79
Q (Contact)	3 Extreme Takes	15.24	15.71	0.91	0.22	0.53
Q (Contact)	Pooled	9.67	12.28	0.62	0.48	0.66
Baseline	–	–	–	*0.92*	–	*0.76*

A. No-Contact Model (Model T)

Model T serves as the baseline for ideal, unloaded kinematic tracking. It was evaluated on three held-out no-contact takes (Take 04, Take 07, Take 28). Because the dataset contains no external loads, the force and contact-state heads are unutilized, and only joint-angle predictions are evaluated.

Across the held-out takes, Model T achieved a mean joint Mean Absolute Error (MAE) of 2.24° (Joint 1: 1.76°, Joint 2: 2.73°). Per-take errors remained narrow, ranging from 1.66° to 3.31°. Without external loads to induce unmodeled material deformation, the remote sensor-to-angle mapping remains highly consistent. The Take 28 joint trajectories are shown in Figure 5, with the model's predicted trajectories compared against the ground truth angles extracted from MOCAP. This graph shows low phase lag and small amounts of steady-state error, most notable for joint 2, which is the distal joint. This movement includes randomly generated poses showing the model can generalize to truly unseen motion.

B. Contact Model (Model Q)

Model Q evaluates the network's capacity to resolve coupled kinematics, force, and contact states. It was trained on a pooled dataset comprising 75 takes across normal contact, no-contact, and extreme hyperextension conditions. It was evaluated on 6 held-out takes: 3 standard contact takes and 3 extreme-loading takes.

Because performance bifurcates sharply based on the physical loading regime, metrics are reported both as an aggregate pool and split by operational condition.

Under standard contact interactions, the model successfully extracts all three modalities from the shared hidden state:

- **Joint Angles:** The model achieves an average MAE of 4.11° (5.09° for Joint 1; 3.13° for Joint 2).

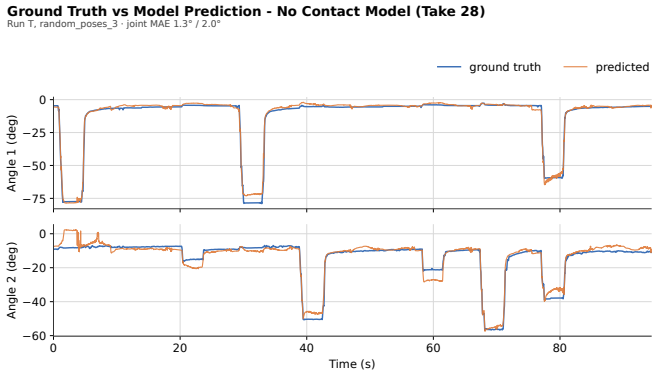


Fig. 5: Model T kinematic tracking performance during a free-space randomized actuation sequence (Take 28, random_poses). Evaluated on this unscripted no-contact test take, the network maintains precise joint estimation across dynamic ranges, achieving an MAE of 1.3° and 2.0° for Angle 1 and Angle 2, respectively.

- **Contact Force:** Force MAE is 0.34 N against a constant-prediction baseline of 0.92 N, with a Spearman rank correlation of 0.73. The model accurately tracks the shape and magnitude of the loading profiles.
- **Contact State:** The classifier reaches an F1 score of 0.79, driven by a precision of 0.84 and a recall of 0.74. The model behaves conservatively, rarely triggering false positives during free-space movement.

This consistent reliability is shown across these test takes: Figure 6 and Figure 7, showcasing the Flex_Hold_100 and Random_Poses movements, respectively. In addition, Figure 8 shows Take 220 with a MuJoCo visualization of the predicted vs ground truth state.

However, Model Q shows reduced performance when subjected to extreme joint conditions and large force impulses. During intentional hyperextension events, the soft finger body undergoes severe, non-linear buckling. In this regime, the mean joint MAE degrades to 15.24°, heavily skewed by a peak error of 45.79° on Joint 1 during Take 102. Force tracking similarly collapses, with the Spearman rank correlation dropping to 0.22.

Consequently, the pooled metrics across all 6 test takes (9.67° joint MAE, 0.62 N force MAE, and 0.66 contact F1) obscure the model's high reliability in normal operating bounds. A summary of the model performance and metrics across the different conditions is listed in Table II.

To properly contextualize these results, the table includes a naive heuristic baseline. The force baseline predicts a constant mean dataset value (0.92 N) to verify the network is learning true dynamic loading rather than simply outputting an average offset. The contact baseline utilizes a trivial "always-on" classifier; because the finger spends significant time pressed against the probe, this heuristic yields an artificially high F1 score of 0.76 through 100% recall. As demonstrated across the tracking trajectories in Figs. 5, 6, and 7, the model captures

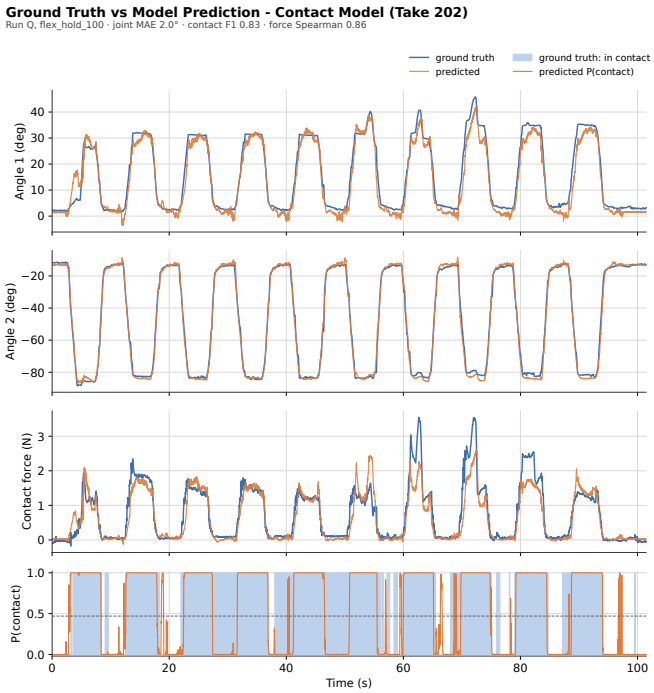


Fig. 6: Model Q tracking performance during a primary flexion movement (Take 202, flex_hold_100). On this held-out test take, the network maintains high-fidelity kinematics (overall joint MAE 2.0°) while accurately resolving rapid contact events (F1 0.83) and dynamic force profiles peaking near 3.5 N (Spearman ρ 0.86).

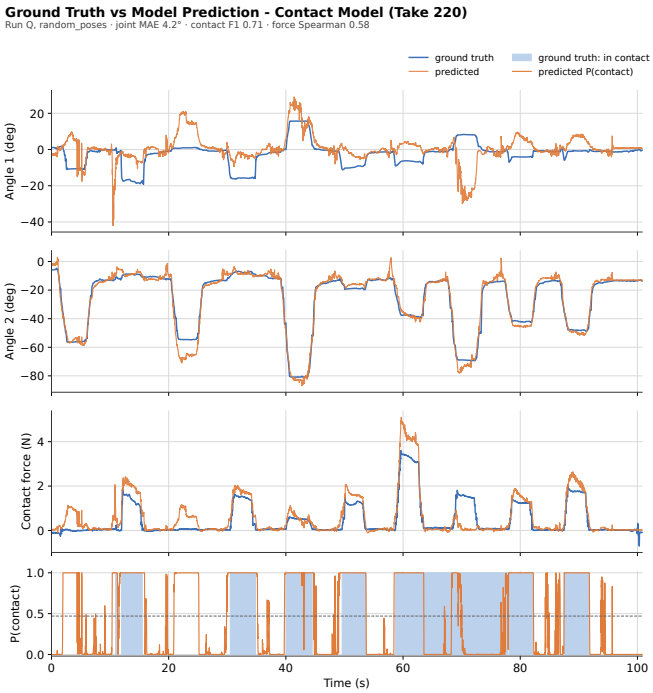


Fig. 7: Model Q tracking performance during a randomized actuation sequence (Take 220, random_poses). Evaluated on this highly variable, unscripted test take, the network maintains continuous tracking (overall joint MAE 4.2°) while resolving irregular contact events (F1 0.71) and complex force profiles peaking near 5.0 N (Spearman ρ 0.58).

dynamic joint variations rather than converging to trivial mean baselines or static heuristics. Comparing Model Q against this baseline reveals that while its standard-take F1 of 0.79 reflects genuine discriminatory ability, its pooled F1 of 0.66 falls below trivial guessing due to the recall collapse on extreme takes. Additionally, the table reports the Spearman rank correlation (ρ) to measure how accurately the model tracks the structural shape of the force profiles independent of scaling errors, alongside the F1-score to evaluate binary contact detection robustly against false positives and false negatives.

VI. Application: Real-Time Contact Force Estimation and Closed-Loop Control of a 2-Finger Soft Hand

We deploy Model Q on the same hardware used for data collection to demonstrate that closed-loop force control based on actuation-side sensing is feasible. The model is trained only on data from the instrumented finger and is deployed to both fingers without retraining. An estimation pipeline buffers motor encoder and tendon-tension data to feed the model, which continuously predicts joint kinematics and contact forces at a fixed cadence without external cameras or embedded sensors.

The resulting real-time force estimate is subsequently routed to a proportional-integral-derivative (PID) controller, which

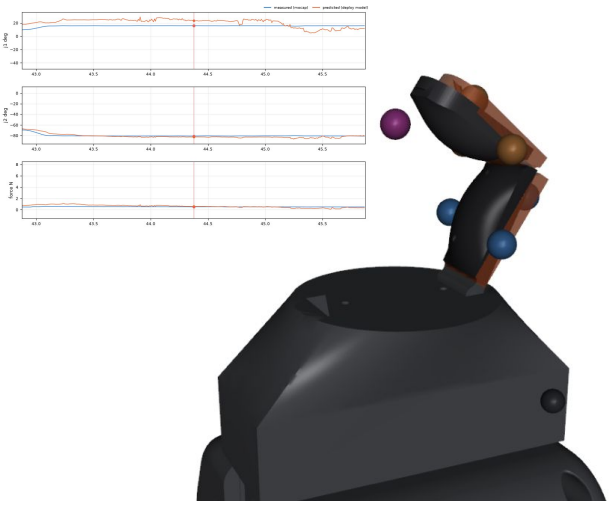


Fig. 8: MuJoCo visualization comparing the ground truth joint state (gray) against Model Q's predicted state (orange) for Take 220. The insert plots display the continuous tracking for joint angles and contact force with the red line indicating the current visualized frame.

calculates tracking errors and actively modulates the commanded joint positions to achieve a target force setpoint. The estimation pipeline runs on a Lenovo ThinkPad P1 Gen

8 workstation laptop. Model Q contains 52,484 trainable parameters (0.21 MB, fp32). Single-frame inference over the full 300-sample window requires a mean of 0.99 ms on the laptop GPU (p95 2.36 ms), which leaves ample margin within the 11 ms period of the 90 Hz control loop. The loop rate is limited to 90 Hz by the buffering path rather than by inference, which occupies approximately 1 ms of the 11 ms period. We validate real-time force predictions against ground-truth measurements obtained by physically pressing the finger against the same external load cell probe utilized during the initial data collection phase. This ensured that the deployed model remained physically grounded and that the control loop would govern based on accurate kinetic feedback. This is shown in Figures 9 and 10.

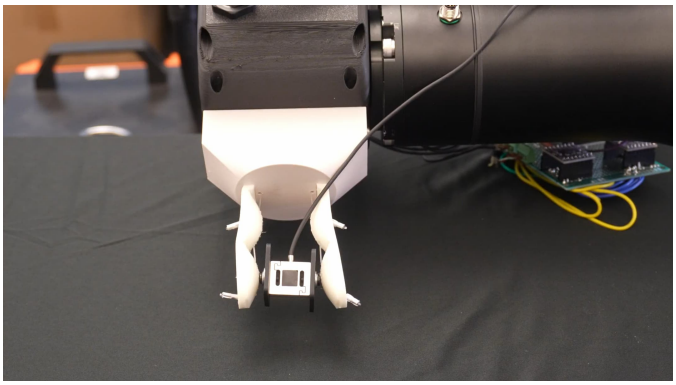


Fig. 9: Hardware setup for real-time closed-loop force control. The hand grasps a custom load cell apparatus for ground-truth validation. The control loop regulates the grip relying entirely on the model's remote force estimates, while the external load cell records the true physical contact force.

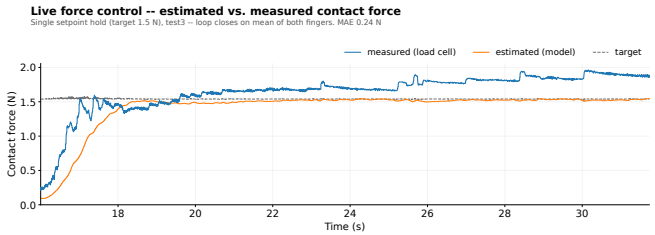


Fig. 10: A segment of the live closed-loop tracking experiment demonstrating the transition from 0 to a 1.5 N setpoint. The controller relies solely on the model's force estimate (orange) to regulate the grip. Ground-truth data from the physical load cell (blue) confirms successful force control with a 0.24 N MAE over this segment. The plot reveals a characteristic discrepancy where the smoothed estimator perfectly tracks the setpoint, while the physical dynamics exhibit minor viscoelastic overshoot and steady-state bias.

Contact forces were inferred at 90 Hz and filtered with an EMA ($\tau = 0.75$ s) before being passed to the controller. This temporal filtering lowered high-frequency estimation noise and stabilized the feedback loop, leading to an overall Mean Absolute Error (MAE) of 0.32 N, with steady-state errors remaining under 0.4 N for contact forces up to 4 N. The model exhibited a slight, load-proportional underestimation (approximately 5–10% of the measured force), where bias grew from -0.06 N at a 2.8 N target to -0.38 N at 4.1 N. This divergence is primarily consistent with the active hinge joint undergoing increased structural deflection at higher loads, though tracking performance can be further improved by fine-tuning the PID gains.

While controlling forces to a fixed setpoint proved highly stable, analyzing the transient step response revealed discrepancies between the predicted state and the true physical dynamics. Following a step-increase in the target force, the physical ground-truth load cell recorded pronounced overshoots, transiently spiking 15–35% above the setpoint, which took 2–3 seconds to fully relax. In contrast, the model's smoothed force estimate ramped cleanly to the target without any corresponding overshoot. As a result, the largest localized prediction errors (peaking at -2.4 N) occurred strictly at these step edges and resolved within 2 seconds. This behavior confirms that the transient discrepancy is an artifact of the low-pass filter's phase lag ($\tau = 0.75$ s) during rapid step transitions rather than intrinsic estimator bias. While this filtering trade-off tailors the current implementation to steady-state and quasi-static force regulation over impulsive impacts, the architecture provides a stable foundation for contact tasks. Future work will explore adaptive filtering and phase compensation to expand closed-loop bandwidth.

VII. Discussion

Looking at the experimental results, an important hardware trade-off question is highlighted: is the integration complexity of tendon-side load cells worth the improved sensing capabilities compared to motor-only models? Introducing load cells into the robotic system increases cost, complicates wire routing, and requires an additional amplification board to process these signals. To isolate the contribution of the tendon-side load cells, motor-only baselines (Models U and V) utilizing 9-channel telemetry were evaluated against their 11-channel, load-cell-equipped counterparts (Models T and Q). A summary of these differences is detailed in Table III.

TABLE III: Model Performance Across Operating Regimes: Motor and Load Cell (MLC) vs. Motor-Only

Model	Evaluation Set	Joint MAE (°)	Norm. Joint Error (%)	Force MAE (N)	Force ρ	Contact F1
T (MLC, No-Contact)	3 Free-Space Takes	2.24	4.21	–	–	–
U (Motor-Only, No-Contact)	3 Free-Space Takes	2.14	4.08	–	–	–
Q (MLC, Contact)	3 Standard Takes	4.11	6.71	0.34	0.73	0.79
V (Motor-Only, Contact)	3 Standard Takes	4.85	7.80	0.41	0.72	0.78
Q (MLC, Contact)	3 Extreme Takes	15.24	15.71	0.91	0.22	0.53
V (Motor-Only, Contact)	3 Extreme Takes	15.17	15.67	0.88	0.08	0.45
Q (MLC, Contact)	Pooled (6 Takes)	9.67	12.28	0.62	0.48	0.66
V (Motor-Only, Contact)	Pooled (6 Takes)	10.01	12.67	0.64	0.40	0.61
Baseline	*Constant / Always-On*	–	–	0.92	–	0.76

In free-space, no-contact motion, both modalities perform identically within training noise boundaries (MAE 2.14° vs. 2.24°), confirming that load cells offer no kinematic advantage when external forces are absent. The current signal in the Dynamixel acts as a proxy for the load cell force and is

sufficient for this non-object-interacting use case. During contact interactions, however, the load cell provides a measurable kinematic improvement of approximately 0.7° across standard loading regimes by directly measuring tendon stretch and material yielding. In the extreme regime, both models degrade to approximately 15° and are indistinguishable (15.24° vs. 15.17°): once the finger enters non-linear buckling, kinematic accuracy is bounded by the out-of-distribution deformation rather than by sensing modality.

The fundamental limitation of the motor-only architecture emerges in force estimation. The absolute force MAE is comparable between modalities (0.62 N vs. 0.64 N pooled), however, the rank correlation, which measures whether the model tracks force variation, highlights the difference (0.48 vs. 0.40 pooled, and 0.22 vs. 0.08 in the extreme regime). Relying solely on drive current as a kinetic proxy, the motor-only model carries little usable force information. Once the mechanically loaded actuator moves toward stall current, current becomes heavily distorted by drivetrain friction and unmodeled compliance. In addition, the motor is unable to track current past a given threshold where the motor becomes overloaded. Consequently, while motor-only sensing is sufficient for free-space kinematics and generalized timing, the dedicated load cell remains necessary to resolve the accurate force structure and magnitudes required for closed-loop regulation.

Beyond sensor modalities, testing revealed two notable system limitations:

- **Structural Buckling:** Evaluating Model Q exposed the model's sensitivity to large deformation. While the model tracks standard multi-DOF interactions with high reliability, performance degrades during extreme, unmodeled hyperextension events where the soft finger undergoes non-linear buckling.
- **Real-Time Bandwidth:** The real-time closed-loop deployment exposed latency inherent to the filtering architecture. Network inference accounts for approximately 1 ms of the 11 ms control period, so the estimator lag originates almost entirely in the EMA filter ($\tau = 0.75$ s) and the telemetry path. The system is highly stable in steady-state but struggles to capture high-frequency physical dynamics. This results in transient overshoots at the physical contact interface during rapid step responses, indicating a need for reduced estimator lag and further tuning of PID and EMA parameters in future high-frequency applications.

VIII. Conclusion

This work extends the learning-based proprioceptive framework proposed in previous work [2] to multi-tendon, multi-degree-of-freedom soft robotic hands. By utilizing a shared-trunk recurrent architecture, the model is able to resolve coupled kinematics, contact onset, and force magnitudes relying exclusively on motor and load cell data. This approach reduces the reliance on external vision systems and advances the feasibility of deploying highly dexterous, visionless soft end-effectors in unstructured environments. Future work will focus on increasing the accuracy, reliability, and repeatability of these model estimations, reducing the transient tracking lag, and expanding the scope of estimation to include granular contact location and rough state estimation of different objects.

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References

[1] A. J. Spiers, A. S. Morgan, K. Srinivasan, B. Calli, and A. M. Dollar, "Using a Variable-Friction Robot Hand to Determine Proprioceptive Features for Object Classification During Within-Hand-Manipulation," *IEEE Transactions on Haptics*, vol. 13, no. 3, pp. 600–610, Jul.–Sep. 2020.

[2] C. Bauer, D. Bauer, and N. S. Pollard, "Learning-based State and Contact Estimation for Tendon-Driven Soft Robot Fingers using Multimodal Sensing," in *2026 9th IEEE-RAS International Conference on Soft Robotics (RoboSoft)*, 2026, pp. 446–453.

[3] S. M. H. Sadati *et al.*, "Stiffness Imaging With a Continuum Appendage: Real-Time Shape and Tip Force Estimation From Base Load Readings," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 2824–2831, Apr. 2020.

[4] K. Gilday, L. Relandeau, and F. Iida, "Design and Characterisation of a Soft Barometric Sensing Skin for Robotic Manipulation," in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022, pp. 10021–10028.

[5] C. Tawk, H. Zhou, E. Sariyildiz, M. in het Panhuis, G. M. Spinks, and G. Alici, "Design, Modeling, and Control of a 3D Printed Monolithic Soft Robotic Finger With Embedded Pneumatic Sensing Chambers," *IEEE/ASME Transactions on Mechatronics*, vol. 26, no. 2, pp. 876–887, Apr. 2021.

[6] K.-Y. Lin, A. Gamboa-Gonzalez, and M. Wehner, "Soft Robotic Sensing, Proprioception via Cable and Microfluidic Transmission," *Electronics*, vol. 10, no. 24, p. 3166, Dec. 2021.

[7] Y. She, S. Q. Liu, P. Yu, and E. Adelson, "Exoskeleton-Covered Soft Finger With Vision-Based Proprioception and Tactile Sensing," in *2020 IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 10075–10081.

[8] X. Liu, X. Han, W. Hong, F. Wan, and C. Song, "Proprioceptive learning with soft polyhedral networks," *The International Journal of Robotics Research*, vol. 43, pp. 1916–1935, 2024.

[9] P. Roberts, M. Zadan, and C. Majidi, "Soft Tactile Sensing Skins for Robotics," *Current Robotics Reports*, vol. 2, pp. 343–354, 2021.

[10] S. Taylor, K. Park, S. Yamsani, and J. Kim, "Fully 3D printable Robot Hand and Soft Tactile Sensor based on Air-pressure and Capacitive Proximity Sensing," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 18100–18105.

[11] P. Weiner, C. Neef, Y. Shibata, Y. Nakamura, and T. Asfour, "An Embedded, Multi-Modal Sensor System for Scalable Robotic and Prosthetic Hand Fingers," *Sensors*, vol. 20, no. 1, p. 101, 2019.

[12] S. Puhlmann, J. Harris, and O. Brock, "RBO Hand 3: A Platform for Soft Dexterous Manipulation," *IEEE Transactions on Robotics*, vol. 38, no. 6, pp. 3434–3449, 2022.

[13] D. Bauer, C. Bauer, A. Lakshmipathy, R. Shu, and N. S. Pollard, "Towards Very Low-Cost Iterative Prototyping for Fully Printable Dexterous Soft Robotic Hands," in *Proceedings of the IEEE International Conference on Soft Robotics (RoboSoft)*, Apr. 2022.

[14] W. Kabsch, "A solution for the best rotation to relate two sets of vectors," *Acta Crystallographica Section A*, vol. 32, no. 5, pp. 922–923, 1976.

[15] K. S. Arun, T. S. Huang, and S. D. Blostein, "Least-Squares Fitting of Two 3-D Point Sets," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. PAMI-9, no. 5, pp. 698–700, 1987.